Research Question

For my project I want to look at the value of "Mega-Rare" items in the game Old School Runescape or OSRS over time as changes have been made to the game. OSRS is a game of incremental upgrades, however these items are designed by the developers to remain at the pinnacle of the game. They are difficult to obtain and because each has remained the "best in slot" for their entire lifespan the price should remain fairly consistent. I want to examine what game updates have had the largest impact on the price of these items and interrogate the reason for price fluctuation over time.

Public Interest

As OSRS is an MMORPG and these items are publicly traded commodities any increased understanding of price change over time could prove valuable to the people who play the game. It is not uncommon for players to treat items the same as real investments posting their gains and losses on online forums like the subreddit /r/GrandExchangeBets.

Hypothesis

My expectation is that these items' prices will be the most heavily impacted in the short term by patches that introduce new "end-game" content. As each of the "mega-rare" items have been added in turn I expect their additions will cause the price of the previously introduced "mega-rares" to change. So far, four of these items have been released and I anticipate a slight reduction in price with each new item.

Sources

API documentation for Runescape wiki:

https://runescape.wiki/w/Application_programming_interface

Endpoints:

https://oldschool.runescape.wiki/api.php,
https://oldschool.runescape.wiki/api.php?action=bucket
https://api.weirdgloop.org/exchange/history/osrs/all

Example Item(*Twisted Bow):*
OSRS site:

https://secure.runescape.com/m=itemdb_oldschool/Twisted+bow/viewitem?obj=20997

Item Search:

https://secure.runescape.com/m=itemdb_oldschool/api/catalogue/items.json?category=1&alpha=twisted%20bow&page=1

Documentation price updates on the wiki:

https://oldschool.runescape.wiki/w/RuneScape:Real-time_Prices

Methods

Right now I have been experimenting with osrs-wrappers found here:

https://pypi.org/project/python-osrsapi/

https://github.com/atjeff/osrs-wrapper

This will give me some historical data for the item prices within specific time frames(see below for an example)

The wiki also has a complete list of game updates with dates here:

https://oldschool.runescape.wiki/w/Game_updates

My current plan is to use requests and beautiful soup to find Game Updates pages that specifically mention my items. I can then create a graph that uses the timestamps for each article to place them over the price over time data. My plan is to use some of the techniques we learned in chapter 10 this week to create strong visual representations of the updates that had the most impact on price and follow that up with an analysis. I think the information in both chapter 4 and 5 will also help when using the beautiful soup with the wiki as they deal with parsing information using xml and json.

The wrapper I have been using has a built-in 30-day window for item price data that I think I'll be able to use to look at the price surrounding these updates. If there is enough activity I could also include smaller charts that deal with a specific update. As there are often many smaller patches following a large one, those smaller ones could affect item prices if they mention the items I am focusing on.

I am planning to collect the price for each item once per day and use that information to compare with the updates. If I find that that is not sufficient I will expand my time window.

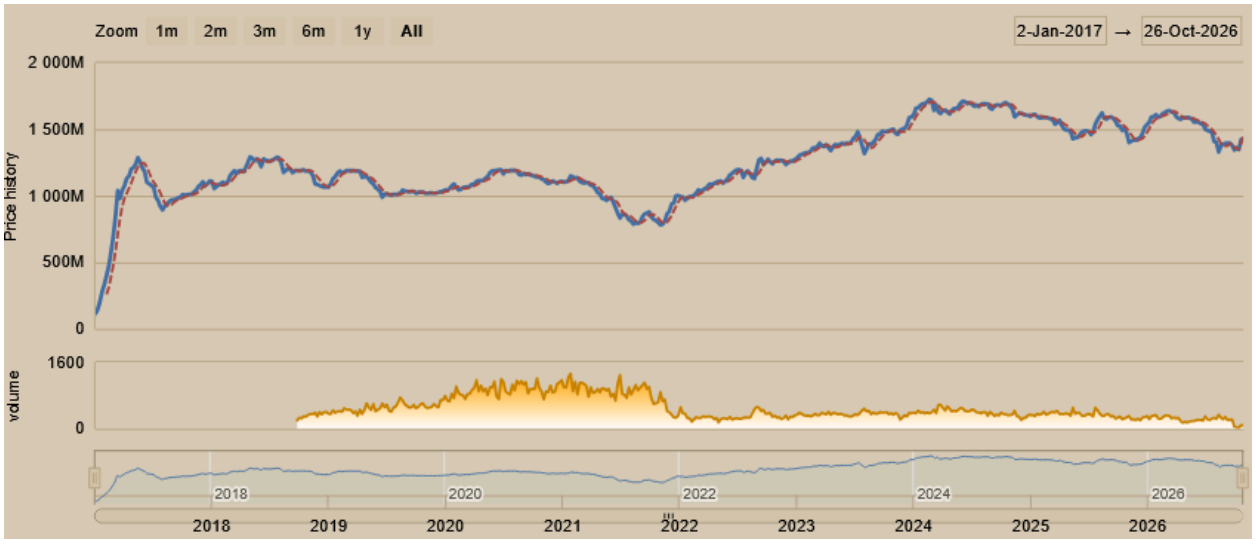
Ethics and feasibility

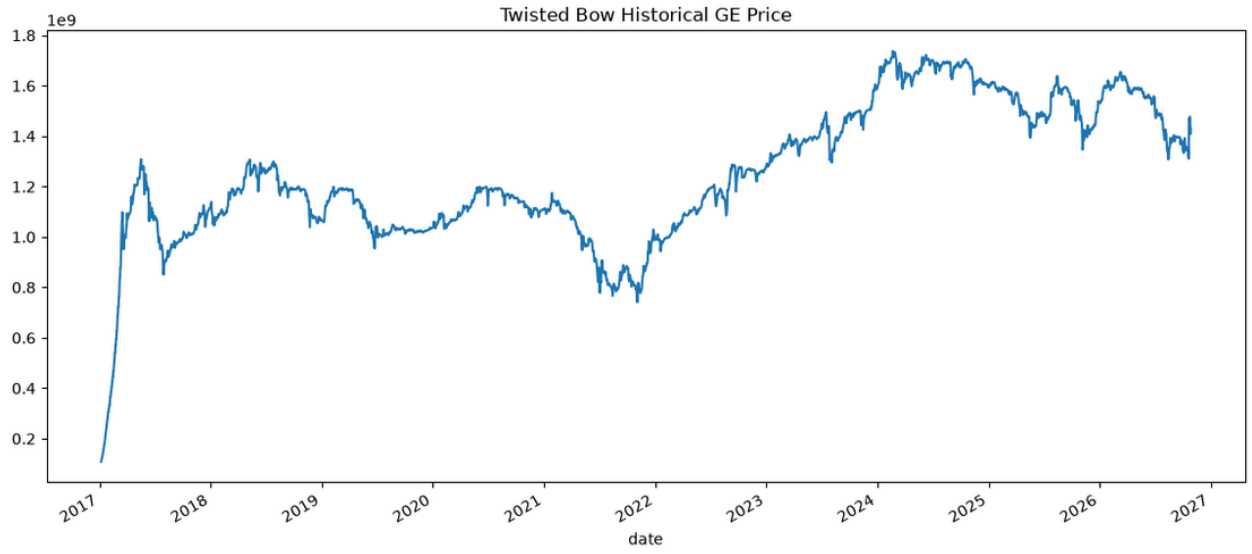
So far I have kept my number of items limited and there is good documentation for everything I am trying to accomplish. Hopefully this will prove to be both achievable and reasonable for my skill level within the timeframe. The wiki has this section regarding its api:

Within reason, we want people to use these APIs as much as they need to build cool projects and tools. We do not explicitly ratelimit any of the endpoints, and we do our best to cache the responses at multiple levels.

However, we reserve the right to limit access to anyone, if their usage is so frequent that it threatens the stability of the entire API. We don't know where that line is right now, but for Grand Exchange prices, it would probably have to be multiple large queries per second for a sustained period. If we end up blocking your tool, feel free to reach out on Discord and we'll see if there's a better solution for what you're doing.

This makes me more confident that so long as I limit my number of requests there will be no issue gathering data using their api. The data I plan to gather won't have any personal data as I am dealing with dates and prices. However, this could change if I see a person/s having an outsized impact on the price.





ChatGPT helped me generate most of methods to change the date range

import requests

import pandas as pd

URL = "https://api.weirdgloop.org/exchange/history/osrs/all"

```
PARAMS = {"id": 20997}

HEADERS = {"User-Agent": "WEBDATASCIENCE OSRS Research Project/1.0,
simon.spillane@colorado.edu"}

response = requests.get(URL, params=PARAMS, headers=HEADERS)

response.raise_for_status()

print(response)

TbowData = response.json()

records = TbowData["20997"]

dfTbow = pd.DataFrame(

    records,

    columns=["timestamp", "price", "volume"][:len(records[0])]

)

dfTbow["date"] = pd.to_datetime(

    dfTbow["timestamp"],

    unit="ms"

)

start_date = "2018-09-26"

end_date = "2026-10-30"


filtered_df = dfTbow[

    (dfTbow["date"] >= start_date) &

    (dfTbow["date"] < "2026-10-01")

]

dfTbow.plot(

    x="date",
```

```
    y="price",
    figsize=(14, 6),
    title="Twisted Bow Historical GE Price",
    legend=False
);
```

Price chart using the wiki for the last year:

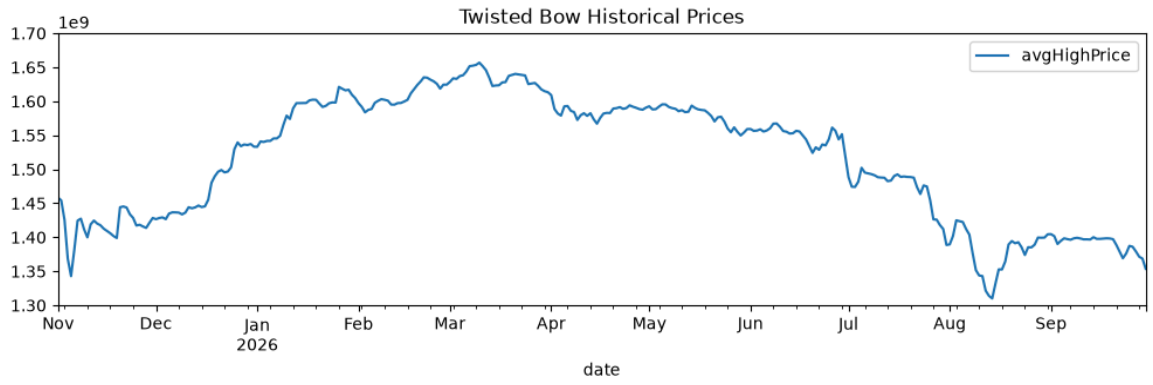
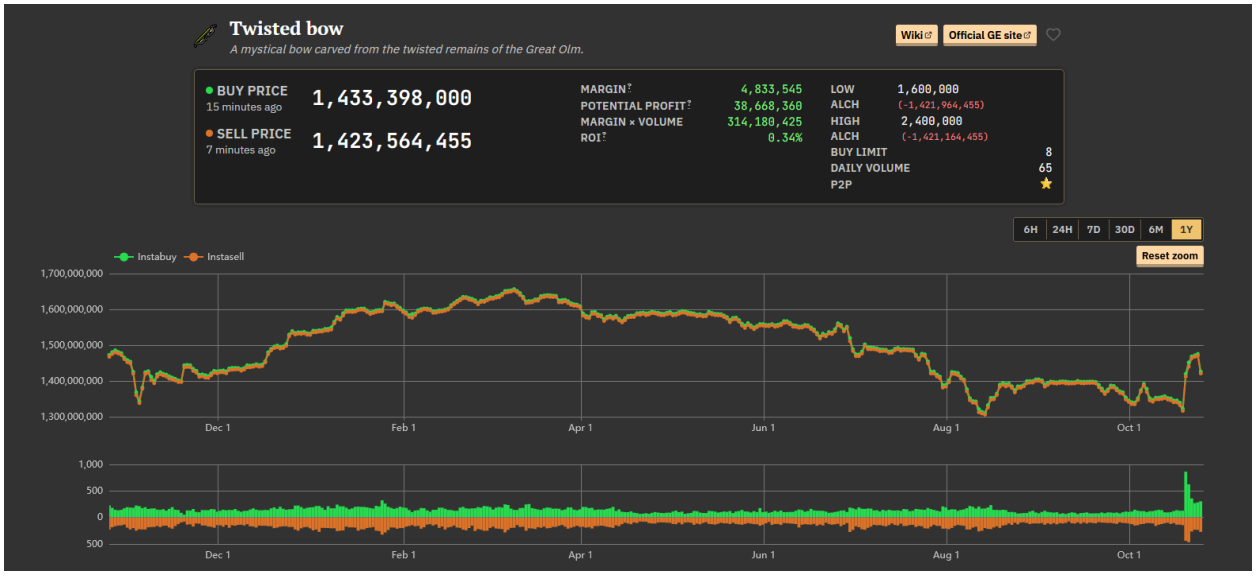


Chart from wiki same timeframe and price range:



This was used to generate the dataframe:

```
import requests
import pandas as pd

item_id = 20997

url = ("https://prices.runescape.wiki/api/v1/osrs/timeseries")

PARAMS = {"id": item_id,"timestep": "24h"}

HEADERS = {"User-Agent": "WEBDATASCIENCE /1.0, simon.spillane@colorado.edu"}

response = requests.get(
   url,
   params=PARAMS,
   headers=HEADERS,
   timeout=30)

response.raise_for_status()

pricedata = response.json()

dfTbow = pd.DataFrame(pricedata["data"])

dfTbow.head()
```

The biggest risk will be my ability to follow through with the project as I haven't done something like this, but I think it's doable within the time frame. I will probably need to rely on some AI models to help me make the graphs as I do not have much practice making them. If my current idea falls through I will try to shrink the scale to a single item or fewer days.